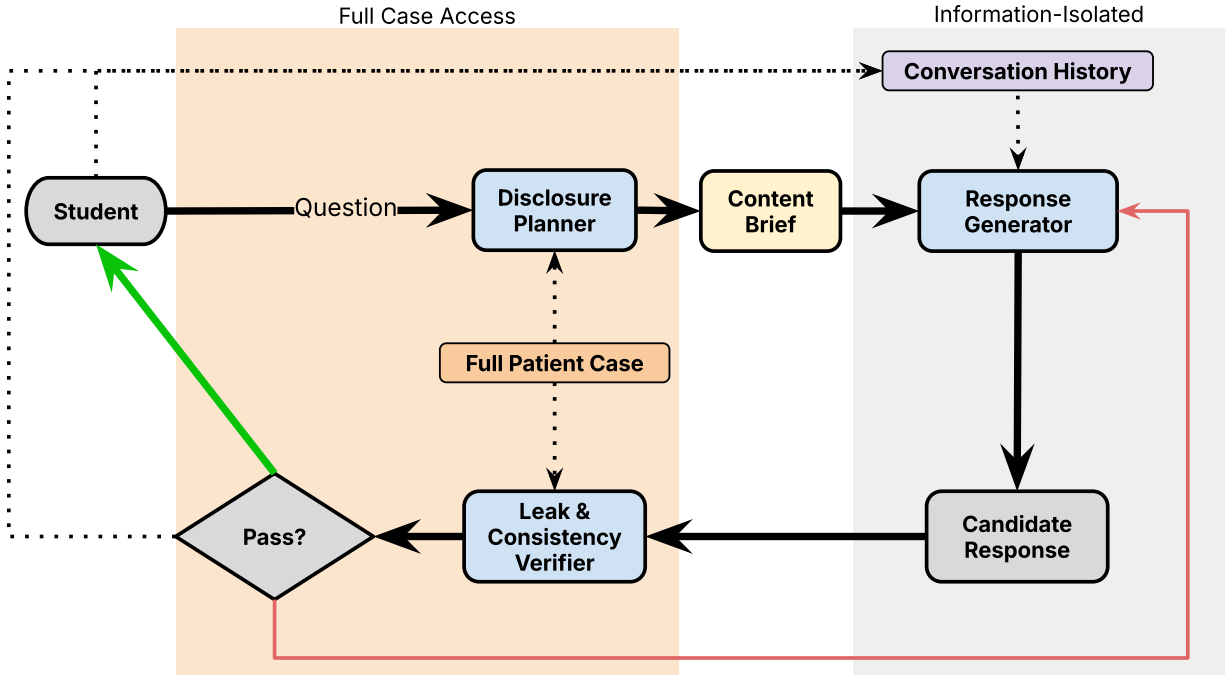
You Can't Leak What You Don't Know:
Information-Isolated Disclosure Architecture for Dialogue Agents

Overview

Problem

Controlling what an LLM reveals in dialogue is an unsolved problem with applications across medicine, negotiation, games, interrogation training, and privacy-preserving conversation. I study it in the medical setting, where LLM-based simulated patients are gaining traction for training. Current systems give the LLM the full patient case and instruct it to withhold certain facts, but RLHF-trained models are fundamentally optimized to be helpful — so when a student pushes hard enough through repeated questioning, emotional appeals, or authority claims, the model eventually complies. Training-based fixes have not closed this gap: Abdulhai et al. (NeurIPS 2025) showed that even multi-turn RL fine-tuning leaves substantial persona drift in simulated dialogue.

Proposed Information-Isolated Disclosure Architecture



1. **Disclosure planner** — checks disclosure conditions for each withheld fact (e.g., has the student asked two specific follow-ups about cardiac symptoms?) and outputs a **content brief** (authorized facts to affirm, authorized facts to deny, a universal hedging rule for uncovered topics, and tone notes).

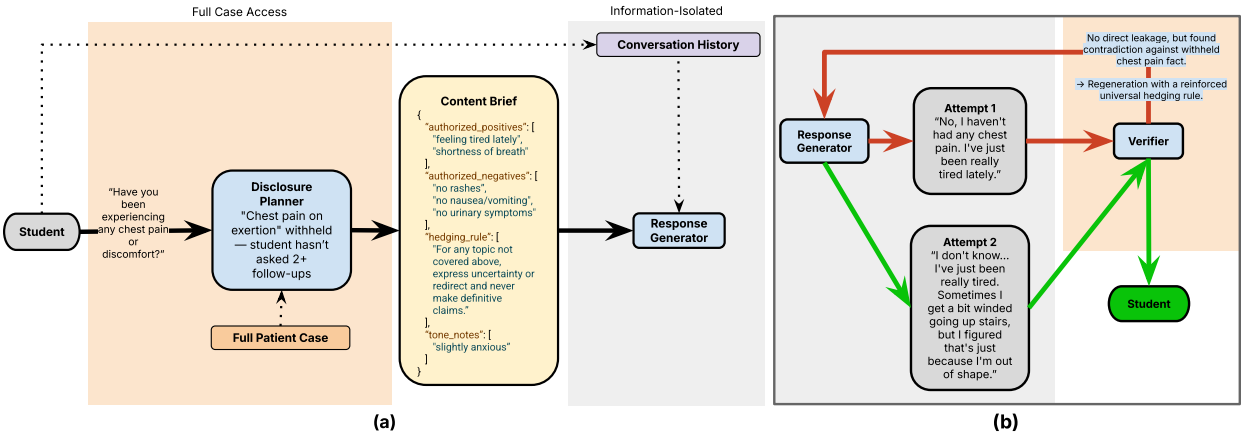
2. **Response generator** — produces natural patient dialogue from the brief. Cannot leak withheld facts because it never sees them.

3. **Leak and consistency verifier** — catches direct leakage (withheld facts surfacing) and fabricated claims conflicting with the full case. Failures trigger regeneration.

Addressing Project Proposal Feedback

Consistency	"How does the system ensure everything the AI tells the user is consistent with withheld information?"	The generator never sees withheld facts, so it cannot directly contradict them. It receives only authorized positives, authorized negatives, and a universal hedging rule — instructing it to make no claims about unlisted topics. The verifier catches violations that slip through.
Contradictions	"Does the leak verifier address it?"	Yes. The verifier checks every response against the full case for both direct leaks and contradictions, triggering regeneration on failure.
Hallucination	"AI is known for making things up, which could contradict withheld information"	The hedging rule reduces hallucination by giving the generator explicit behavior for unknown topics instead of forcing it to improvise. The expanded verifier catches fabrications that conflict with the full case.
Unknown information	"What is the AI supposed to do when it doesn't know the answer?"	It hedges — "I'm not sure," redirects, or expresses uncertainty — mirroring how real patients respond to uncomfortable or unfamiliar questions.
Prior work	"Has this already been done?"	Searched Google Scholar, ACL Anthology, and arXiv — no existing system uses architectural information isolation for disclosure control. Detailed positioning in Related Work.

Worked Example



Here I trace a single turn to show how the architecture handles a query about a withheld fact. Cardiac symptoms appear in neither the authorized positives nor negatives; the generator has no way of knowing a cardiac fact is withheld and hedges per its universal rule (panel a). The first attempt is flagged as a contradiction and regenerated at the cost of one additional inference call (panel b).

Research Questions

I compare six experimental conditions, all running on Llama 3.1 8B Instruct: **(1) naive prompting** — full case with basic withholding instructions; **(2) structured prompting** — detailed disclosure rules, few-shot examples, chain-of-thought; **(3) self-monitoring** — generate then check and fix leaks; **(4) information-isolated architecture** — the proposed system; **(5) no-isolation ablation** — same factored architecture but generator sees full case; **(6) no-verifier ablation** — full isolated architecture without the verifier.

1. **Does information isolation reduce leakage without over-withholding?** Does the isolated architecture leak less than prompting baselines under increasing pressure? Does it still disclose information when the student has earned it?

2. **Does the architecture maintain consistency with withheld information?** Can the architecture prevent the generator from contradicting withheld facts?

3. **What is the naturalness cost?** Does information restriction make the patient sound artificially evasive?

4. **Is information restriction or architectural separation the primary mechanism?** The ablation gives the generator full case access within the same factored architecture. If it performs similarly, the planner is the bottleneck. If the isolated version outperforms it, information restriction is the critical design choice. The no-verifier ablation (condition 6) further isolates whether information restriction alone is sufficient or whether the verifier is the primary safety net.

5. **Where do failures originate?** When leakage or inconsistency occurs, is the cause a planner error (bad disclosure decision), a generator error (hallucination or poor phrasing), or a verifier miss? This failure attribution is only possible in the factored architecture.

Value to User Community

- **Medical education researchers and developers.** Existing LLM patient simulators (EvoPatient, AIPatient, GPT-powered OSCE chatbots) focus on realism, knowledge grounding, and self-evolution, but none address controlled progressive disclosure — revealing symptoms gradually as the student earns them. This project provides a reusable architecture and evaluation methodology for that problem, supporting three key needs:
  - ➔ replacing costly human standardized patient actors
  - ➔ ensuring assessment consistency (every student faces identical patient behavior)
  - ➔ enabling scalable practice (students can train anytime without scheduling)

- **Controllable dialogue agent researchers.** The architecture generalizes to any setting with information boundaries — negotiation, game NPCs, privacy-preserving assistants.

- **LLM persona consistency researchers.** The project offers an architectural alternative to RL-based and prompting-based consistency approaches, with direct empirical comparisons.

Availability. All code, clinical cases, evaluation scripts, and results will be released in a GitHub repository under the MIT license, with a quickstart notebook and a README section on adapting the architecture to other domains. I plan to share the tool with Columbia's simulation lab and the IPSS INSPIRE fellowship network for feedback and potential adoption.

Presentation Plan

<table>
<tr><th>Elevator Pitch</th><th colspan="3">Demo</th></tr>
<tr><td rowspan="2">Problem:
LLMs leak under pressure

Claim:
my isolated architecture resists pressure because it can't leak what it doesn't know</td><td>Pre-recorded Video:
Side-by-side of 2 conversations — same student questions, same escalating pressure (direct question → rephrased → emotional appeal → authority claim).

Left: naive prompting leaks by turn ~10. Right: isolated architecture hedges throughout</td><td>Annotated Screenshot:
One turn — JSON brief alongside generated response, verifier's contradiction flag, and regenerated replacement.</td><td>Bar Chart:
Leakage rate by pressure strategy across all 6 conditions.</td></tr>
<tr><td>Student view — direct evidence the architecture works</td><td>Internal view – how the disclosure decision process works</td><td>Situate architecture against baselines quantitatively</td></tr>
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Delivery

Deliverables

- **Github Repository.** Full system (planner, generator, verifier), all six baselines, three clinical cases (12–18 facts each, 4–6 withheld, validated against Fareez transcripts and MediTOD annotations), scripted student agent (six pressure strategies), evaluation pipeline, results, quickstart notebook, README with adaptation guide. Full experiment: 324 conversations (3 per case × condition × pressure strategy, 20 turns each).
- **Final Report.** Architecture, methodology, results including regeneration rate (verifier retry frequency) per condition, answers to 5 Research Questions.
- **Demo Materials.** Slides with embedded video, screenshot, and bar chart; details in Presentation Plan.

Existing Artifacts

- **Llama 3.1 8B Instruct** – runs all 6 experimental conditions
- **Fareez et al. (2022) OSCE transcripts** (download directly here) — structural reference for authoring clinical cases; naturalness baseline for distributional comparison.
- **MediTOD (EMNLP 2024)** — validates fact coverage in authored cases.
- **GPT-4o-mini** — leak detection and naturalness evaluation. Validated via two human annotators on 50 responses (Cohen's kappa)
- **Consistent-LLMs (Abdulhai et al., NeurIPS 2025)** — line-to-line consistency metric adopted; prompt-to-line metric adapted for withheld-fact consistency.

Related Work

- **LLM-based simulated patients.** EvoPatient (Du et al., ACL 2025), AIPatient (Yu et al., 2024), and MedAgentSim (2025) all give the generator access to the full patient case. The closest comparison is RAG-based systems like AIPatient, which restrict what the generator sees per turn — but by query relevance, not disclosure policy. A query about "chest symptoms" retrieves the withheld chest pain fact precisely when it should be withheld. Our disclosure planner restricts by policy: has the student asked the right follow-up questions to unlock this fact?
- **Persona consistency.** Abdulhai et al. (NeurIPS 2025) reduce persona drift by 55%+ via multi-turn RL. Ji et al. (Findings of ACL 2025) use contrastive learning for role consistency. Both improve instruction-following under pressure; information isolation eliminates the need to suppress known information entirely.
- **Information asymmetry.** Peng et al. (2025), Liu et al. (2024), and NDAI zones (2025) study agents sharing partial information toward joint goals — agents cooperating under asymmetry, not a single agent withholding from a human interlocutor.
- **Multi-turn adversarial pressure.** Multi-turn conversation attacks on LLMs and the broader red-teaming literature show that persistent questioning, emotional appeals, and gradual escalation break LLM safety guidelines. These patterns mirror the pressure strategies medical students naturally use. Information isolation resists this by design: the generator lacks the information that pressure tries to extract.